

Self-reconstruction of C-point beams

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Abstract: C-point beams are reported to reconstruct themselves on propagation after being obstructed by an opaque symmetric triangular obstruction. The component scalar vortices of this polarization singular beam manage to maintain a relative phase and intensity value to regain their polarization, intensity, and Stokes phase structure. This self-healing property of vortex beams is related to the Poynting vector rotation of such beams. This property of the C-point can be used for meaningful applications in the field of robust optical communication and microscopy. © 2022 The Author(s)

1. Introduction

The superposition of two circularly polarized, orthogonal beams, each having a distinct vortex, can produce a beam with polarization singularity [1]. The helical phase difference between the beams in addition to the amplitude difference, makes the polarization distribution of the beam uneven. This implies that the polarization of the beam has a varying spatially distribution. There are singularities existing in these inhomogeneous polarization structure known as C-points, V-points, and L-lines. C-points are points of circular polarization with elliptical polarization surrounding it with azimuth varying in a particular fashion. These singularities are superposition of both orbital and spin angular momentum. These beams can be expressed as [1, 2],

$$\vec{E}(r, \phi) = [r^{|m|} \exp(im\phi) \hat{e}_R + r^{|n|} \exp(in\phi + \phi_0) \hat{e}_L] * \text{circ} \left\{ \frac{r}{R} \right\}, \quad (1)$$

where, m and n are the charges of the individual vortices. C-points are generated for $m \neq n$ with index given by $I_C = (n - m)/2$. The phase singularities in Stokes field S_{12} is related to these polarization singularities. C-points are claimed to be more stable and robust optical beam compared to any other complex structured light beams [2]. Polarization degree also adds an extra degree of freedom which can be efficiently utilized in several applications like optical communication. We report the self-reconstruction of these C-points on propagation after getting obstructed, for the first time.

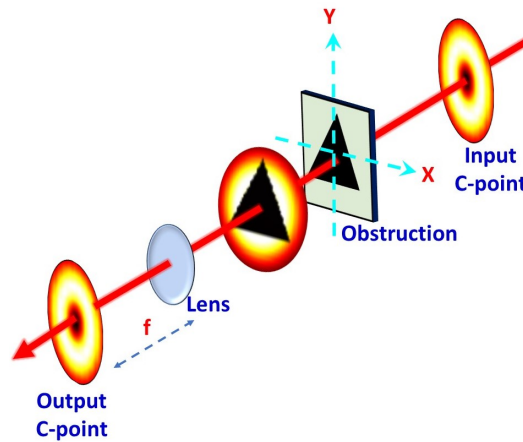


Fig. 1. Schematic showing the self-reconstruction of a C-point at far-field when obstructed by a triangular obstruction.

2. Theory

The scalar optical vortex beam has helical wavefront with its Poynting vector also rotating in helical direction [3,5]. Since Poynting vector depicts the direction of energy flow, the rotation of this vector in azimuthal direction plays a significant role in reconstruction of the beam in its azimuth direction. The bright C-points are generated with the superposition of a Gaussian beam with a scalar vortex beam. In a bright C-point, the scalar vortex beam heals due to this helical Poynting vector rotation. On the other hand, the Gaussian component also interestingly heals itself simultaneously [4]. Therefore, there is simultaneous healing of both the right and left circularly polarized component, which ensures the complete healing of the C-point.

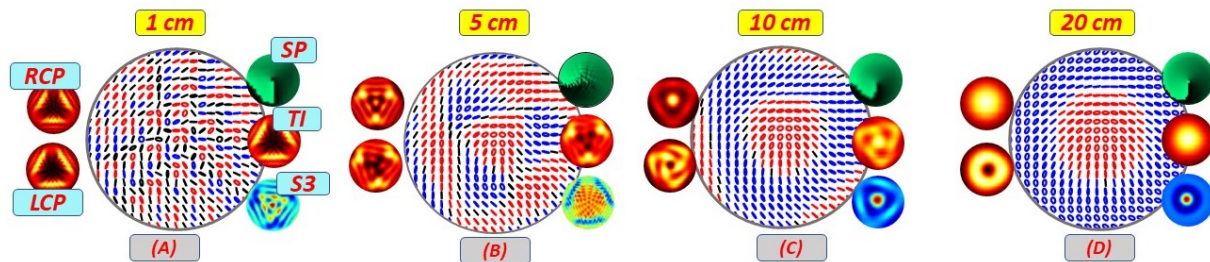


Fig. 2. Self-reconstruction of C-points on propagation. The polarization distribution of the disturbed C-point distribution at different distances: (A) 1 cm, (B) 5 cm, (C) 10 cm, (D) 20 cm (far-field). Right handed component intensity (RCP) and left-handed component intensity (LCP), Stokes phase S_{12} (SP), total intensity (TI), and Stokes parameter S_3 is also shown at each distance.

The simulation result depicting this claim is shown in the Fig. 2. The input C-point beam of index $I_c = +1/2$ is passed through a triangular opaque obstruction. The beam is totally disturbed with distorted intensity (both individuals and total), and polarization distributions immediately after the opaque obstruction as can be seen in Fig. 2 (A). But, as the beam approaches the far-field, the scalar vortex component (here, LCP) and the Gaussian component (here, RCP) heal themselves. This can be witnessed in the individual intensity patterns clearly. The original C-point structure as a whole tries to regain itself in the exact form as in the input in the far-field (Fig. 2 (D)). The relative phases and amplitude in the individual components are maintained to reconstruct its polarization distribution. This holds as a promising property of these beams where it can be used as a promising candidate for optical communication in free-space and microscopy [7]. Stokes phase (ϕ_{12}) distribution and Stokes parameter S_3 also assures this healing.

3. Conclusion

A bright C-point is reported to self-reconstruct on its propagation in free-space after being obstructed for the first time. The individual circularly polarized Gaussian and scalar vortex component heal themselves to reconstruct the overall structure. Polarization distribution is also regained to almost its original form which makes it an interesting and promising candidate for several possible applications.

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